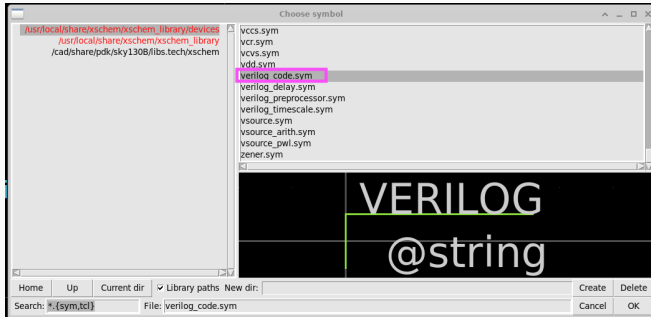


EE5311: Digital IC Design

Tutorial 9

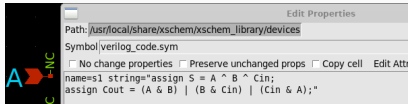
Experiment 9.1

Create a schematic for the full adder verilog module using the `verilog_code` symbol:

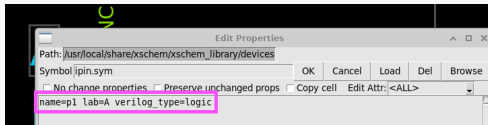


Experiment 9.1 - full adder module

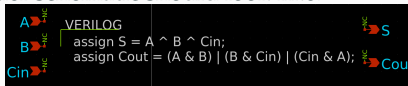
- Populate the verilog code for the full adder:



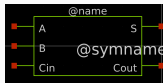
- Add the input and output ports with the names used in the code
- Add pins and modify its type to logic using verilog_type=logic:



- Connect ports to noconn block to prevent xschem's undriven net error
- The final full adder schematic should look like:

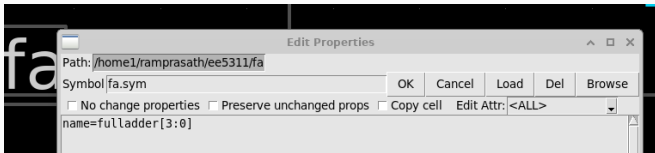


- Create a symbol for the full adder block:

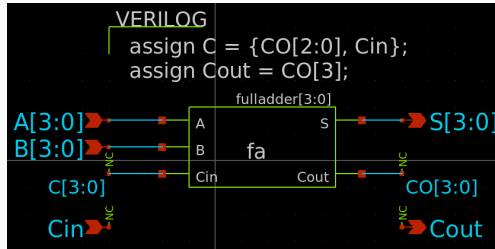


Experiment 9.1 - A 4-bit ripple carry adder

- Create an array of four full-adders in xschem. Arrays are automatically created if you label the instance:



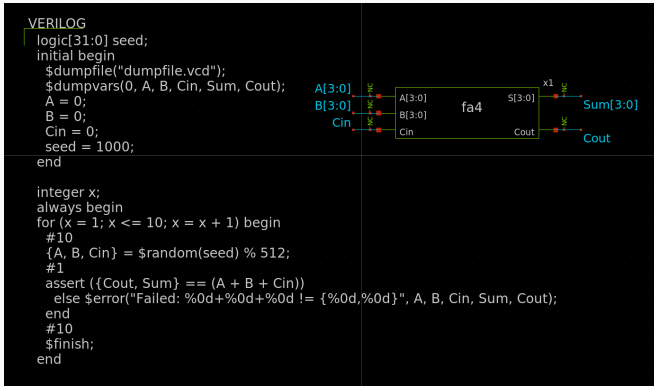
- Add ports and make appropriate connections for the 4-bit adder:



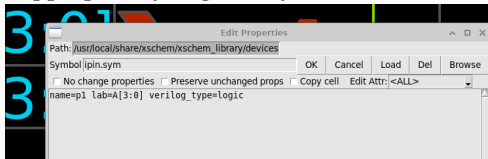
- Create a symbol for the ripple carry block:

Experiment 9.1 - testbench

- Create a testbench for the 4-bit ripple carry adder:

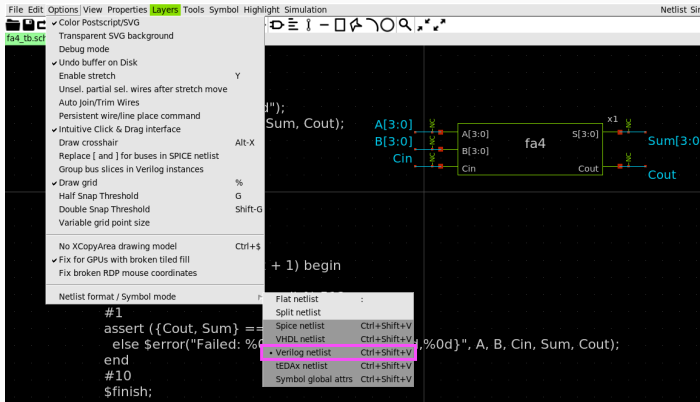


- Label the pins appropriately to get arrays/buses:



Experiment 9.1 - netlist mode

- Change the netlist mode to Verilog using Input and output pins are in Options ► Netlist format ► Verilog netlist:



- Netlist and Simulate
- Open the waveform in the external viewer gtkwave